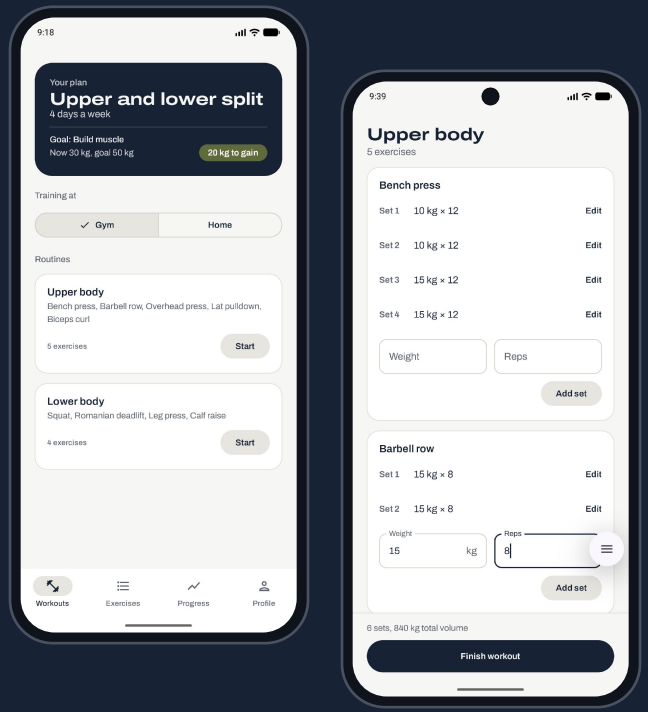
CS 501 · PROJECT UPDATE 1 · NOVEMBER 10, 2026

RepTrack

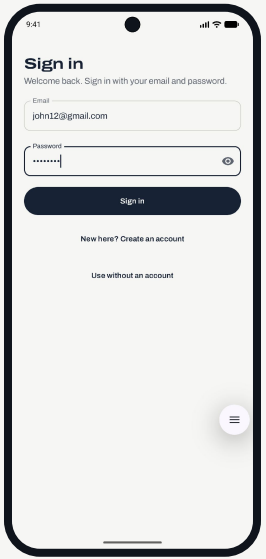
A working slice: sign in, set your goal, log a workout, and see it saved.

Aayush Kumar · Patrick Kola

GitHub

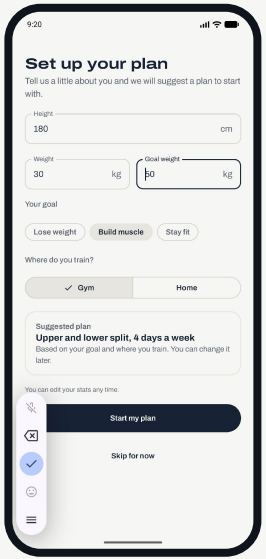


The Nov 10 slice works end to end



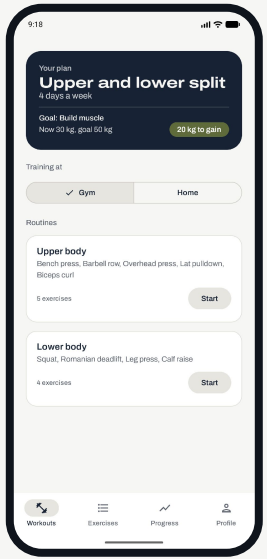
1 · Sign in

Firebase email account



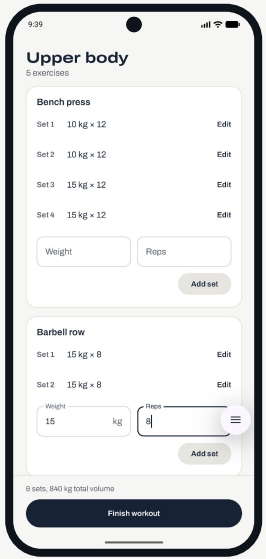
2 · Goal setup

Saved in DataStore



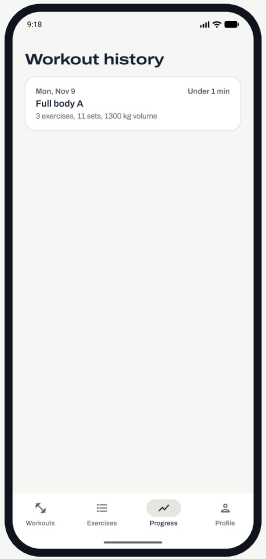
3 · Workouts

Plan from your goal



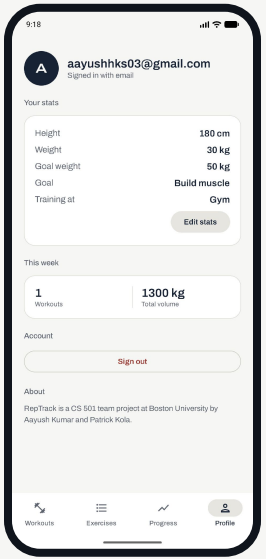
4 · Log sets

Saved to Room every set



5 · Progress

History from Room



6 · Profile

This week, sign out

LIVE DEMO

Architecture: built and planned

	COMPOSE SCREEN	→ VIEWMODEL	→ REPOSITORY	→ DATA SOURCE	STATUS
Account	Sign in	→ SignIn VM	→ Auth repository	→ **Firebase Auth**	**Built**
Profile and goal	Goal setup, Profile	→ GoalSetup, Profile VMs	→ Profile repository	→ **DataStore**	**Built**
Workouts	Workouts, Session, Progress	→ Workouts, Session, Progress VMs	→ Workout repository	→ **Room, single source of truth**	**Built**
Exercise catalog	Exercises tab, swaps	→ Exercises VM	→ Exercise repository	→ **wger API via Ktor, cached in Room**	**Nov 24**
Runs (GPS)	Run tracker	→ Run VM	→ Run repository, location tracker	→ **Fused Location, foreground service**	**Nov 24**
Weight trend	Chart on Progress	→ Progress VM	→ Profile repository	→ **Room weight log**	**Nov 24**
Cloud backup	Backup status on Profile	→ Profile VM	→ Backup worker (WorkManager)	→ **Cloud Firestore**	**Dec 8**

One tap, end to end: logging a set

#	What happens	Where
1	You tap Add set; the card only reports the tap up	WorkoutSessionScreen.kt:275
2	The ViewModel checks weight and reps; bad input shows an error and saves nothing	WorkoutSessionViewModel.kt:128 SetInput.kt:36
3	A coroutine starts; a Mutex creates the session once and keeps its id in SavedStateHandle	WorkoutSessionViewModel.kt:137, :217
4	One Room @Transaction numbers the set and inserts it	RoomWorkoutRepository.kt:36 WorkoutDao.kt:98
5	Room's Flow sends the new list of sets	WorkoutDao.kt:49
6	The ViewModel rebuilds one UI state with flatMapLatest and combine	WorkoutSessionViewModel.kt:105–115
7	The screen collects it and Compose redraws the new row	WorkoutSessionScreen.kt:71

Persistence and tests

Room · workouts and sets

- Tables: workout_sessions, workout_sets
- CASCADE foreign key, session_id index
- Transactions: add, delete, finish
- History: one LEFT JOIN query

DataStore

Height, weight, goal weight, goal, gym or home, setup done

Firebase Auth

Email sign-in and new accounts; the password is never saved to disk

100

unit tests: ViewModels with fake repositories, validators, DataStore

38

Compose UI and Room tests on the emulator

4

tests that typed input and the workout come back after the app is killed

Done, next, and what changed in scope

Done for Nov 10

- Email sign-in and accounts
- Goal setup in DataStore
- Workouts tab, starter plans
- Log, edit, delete sets in Room
- History on Progress
- Profile, weekly totals, sign out

Next: Nov 24 beta

- GPS run tracker (Aayush)
- wger catalog, Ktor (Patrick)
- Weight log and trend chart
- Rest timer between sets
- Accessibility and error states

Later: Dec 8

- Swap an exercise
- Firestore backup with WorkManager
- Sign in with Google

Scope changes to approve

Ktor instead of Retrofit for wger · Google sign-in moves to P1 · new Profile tab for account and stats.

Prioritized backlog

Scrum master this week: Aayush

Priority	Story: as a user, I can…	Owner	Target	Status
P0	Create an account and set my goal	Both	Nov 10	Done
P0	Log and resume a workout	Both	Nov 10	Done
P0	Track a walk or run with GPS	Aayush	Nov 24	Next
P0	Browse the exercise catalog from wger	Patrick	Nov 24	Next
P0	Log my weight and see my trend to goal	Aayush	Nov 24	Next
P0	Rest timer between sets	Aayush	Nov 24	Next
P0	Accessibility and error-state pass	Both	Nov 24	Next
P1	Swap an unavailable exercise	Patrick	Dec 8	Later
P1	Back up and restore my data	Patrick	Dec 8	Later
P1	Sign in with Google	Patrick	Dec 8	Later

Repository record

8
pull requests merged, one branch per feature

0
direct pushes to main since setup

10
issues on the board, labeled P0 or P1

CI
build and unit tests on every PR; main is green

Aayush · 7 PRs

App shell and navigation, theme, goal setup with DataStore, Workouts tab, workout session screens and ViewModel, Progress, Profile, back navigation

Patrick · 1 large PR

Room database: entities, DAO, repository and Room tests.
Firebase sign-in: screen, ViewModel, repository and tests.
Firebase config as a CI secret

Risks and blockers

No blockers today. Solved this sprint: Firebase config in CI, UI tests.

PRIVACY

Run routes

Routes never leave the phone; only totals get backed up. Firebase holds just the account.

PERMISSIONS

Location and notifications

Asked when you tap Start run, with a reason. If denied, the rest of the app still works.

API

wger down or incomplete

Cache the catalog in Room; loading, empty and error states; type an exercise by hand.

DEVICE

GPS accuracy and battery

Drop low-accuracy points, run GPS only during a run, test with emulator mock routes.

USABILITY

Logging between sets

Numbers stay filled for the next set, 48 dp targets, TalkBack labels.

SCHEDULE

Five P0s in two weeks

Split by owner, check in twice a week, cut P1s first if we slip.

Generative AI use

- Used AI to improve code readability through formatting, comments, and documentation.
- Leveraged AI to brainstorm ideas, explore potential solutions, and evaluate different approaches.
- Used AI assistance to debug code, identify errors, and troubleshoot implementation issues.
- Used AI to identify potential edge cases that we might have overlooked.